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Relationship between freshwater scarcity, rural livelihoods and coping strategies in Jirapa Municipality, Ghana

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Abstract

Rural households increasingly rely on surface and groundwater resources, placing pressure on both the quantity and quality of the limited freshwater in Northern Ghana. This study examines the relationship between freshwater scarcity and rural livelihoods, and analyse the coping strategies of households in Jirapa Municipality in the Upper West Region of Ghana. Underpinned by pragmatism, the research adopts a mixed-methods approach to collect data from 172 respondents. Quantitative data were analysed using descriptive statistics, Pearson Chi-square tests, and Likert-scale mean score analysis, while qualitative data were analysed using descriptive narratives to contextualise household experiences and adaptive responses. The findings show that 54.6% of households depend on boreholes as their primary source of water, while 74% of them travel more than 500 m to access water, indicating substantial time and labour burdens. Inferential analysis reveals a statistically significant association between water scarcity and socio-economic livelihood outcomes ($X^2 = 21.073$, $P = 0.001$), with 91.3% of households under severe water scarcity reporting poor livelihood conditions. Perception-based analysis further indicates that water scarcity strongly contributes to poverty and hunger (mean = 4.84), reduces agricultural productivity (mean = 4.73), and increases women's workload (mean = 4.65), while affecting health, sanitation, and children's education. Households adopt various coping strategies, including rainwater harvesting (100%), water rationing (97%), and short-term water storage (89%), although these strategies remain largely reactive and insufficient for long-term resilience. The study concludes that freshwater scarcity is a multidimensional livelihood stressor requiring integrated, context-specific water governance and livelihood support interventions to enhance rural resilience.

Keywords: Freshwater scarcity, Rural livelihoods, Coping strategies, Rural communities, Ghana

1 Introduction

Access to sufficient and safe freshwater is essential for sustaining livelihoods, supporting food production, maintaining health, and ensuring socio-economic development. Globally, water scarcity has emerged as a critical challenge, exacerbated by population growth, climate variability, and competing sectoral demands [1]. Over 2.3 billion people live in water-scarce countries, of whom 733 million are in high or critically water-scarce regions; 1.42 billion people, including 450 million children, face high or extreme water vulnerability [1, 2]. About 4 billion people experience severe water scarcity for at least one month annually [3]. In semi-arid and arid regions of Africa, such as Northern Ghana, smallholder farmers and rural households increasingly rely on limited surface and groundwater resources, placing pressure on both the quantity and quality of water available and threatening the sustainability of livelihoods [4].

Water is deeply embedded in the social and cultural fabric of many rural communities, shaping traditional practices, festivals, and rituals that are essential for community cohesion and identity [12–14]. Water scarcity exerts profound socio-cultural, economic, and gendered impacts. The depletion of rivers, streams, and other water sources undermines these practices, eroding cultural heritage and weakening social bonds. Wellington et al. [15], for instance, postulate that limited water availability affects hygiene and sanitation, disproportionately burdening women and children. In many rural African communities, women spend several hours daily collecting water, often travelling long distances under physically demanding and unsafe conditions [16]. This has cascading consequences for household health, school attendance, and economic productivity, particularly for girls who may miss school or underperform academically due to water-related chores [2]. Water scarcity also influences marital and household dynamics, as women's time-intensive water collection responsibilities reduce their availability for other family and livelihood activities, sometimes generating household conflicts [17, 18].

Agriculture in sub-Saharan Africa is predominantly rainfed, with approximately 95% of crop production dependent on seasonal rainfall [20]. Economically, water scarcity undermines the productivity and income of smallholder farmers, who constitute the majority of food-insecure populations in rural Ghana [19]. Droughts and irregular rainfall patterns reduce crop yields, dry up rivers, lakes, and small dams, disrupt livestock rearing, and threaten food security and income generation [16]. Water-dependent non-farm activities, including petty trading, shea butter processing, and *pito* brewing, are also constrained, particularly for women, further entrenching poverty and limiting investment in high-return livelihoods [21]. Consequently, heads of households usually migrate to urban or other rural areas in search of alternative income sources, affecting local labour availability and social cohesion [22].

Freshwater scarcity in Ghana, particularly in the northern regions, is increasingly a localized and seasonal challenge despite the country's overall water-endowed status. Disparities in access to safe and reliable water persist across ecological zones, with rural northern populations most affected due to climatic variability and limited infrastructure [2, 5, 6]. While national coverage of basic drinking water services has improved, the gains often obscure challenges related to reliability, distance, and seasonal availability, particularly in rural areas where groundwater remains the dominant source [7]. Extant studies highlight that borehole dependence in northern Ghana is increasingly strained by declining groundwater tables, frequent breakdowns, and rising demand, exacerbated by climate variability and population pressures [6, 8, 33]. The Upper West Region, including Jirapa Municipality, represents one of the most water-insecure zones in Ghana, characterized by a unimodal rainfall regime, prolonged dry seasons, and high inter-annual rainfall variability. Recent climate assessments indicate that northern Ghana is experiencing increasing temperature trends and erratic precipitation patterns, which are reducing groundwater recharge and intensifying seasonal water scarcity [9, 10]. In Jirapa Municipality, where over 60% of households depend on rain-fed agriculture, these hydrological constraints directly translate into livelihood vulnerability, reduced agricultural productivity, and heightened gendered burdens associated with water collection [5, 11]. Empirical evidence further suggests that households in similar semi-arid districts adopt short-term coping strategies that are often insufficient to build long-term resilience, thereby reinforcing cycles of poverty and environmental stress [1].

To mitigate these impacts, rural communities employ diverse coping strategies. Rainwater harvesting, for instance, is widely adopted during rainy seasons, providing water for domestic and agricultural use [23]. Also, water storage in tanks, drums, and reservoirs ensures water availability during dry periods [24], while reliance on water vendors supplements household supply, albeit often at higher financial and quality risks [25, 26, 33]. Households may also reduce water consumption or employ water pumping technologies to access distant or underground sources, including manual, solar, and wind-powered pumps [25, 27]. In many rural communities, the dependence on unsafe surface water sources like streams, ponds, and dams persists due to the absence of modern infrastructure, creating both health and reliability concerns [8].

Despite substantial evidence on the impacts of water scarcity, several critical knowledge gaps remain. Few studies have comprehensively examined the integrated socio-cultural and economic dimensions of freshwater scarcity in Ghana, particularly in rural northern contexts such as Jirapa Municipality [29, 30]. Existing research has often focused on isolated aspects, neglecting how household-level coping strategies and local socio-cultural practices intersect to influence livelihoods. Furthermore, the linkages between water scarcity and broader sustainable development objectives, particularly the United Nations Sustainable Development Goals (SDGs) 2 (Zero Hunger); 3 (Good Health and Well-being); 5 (Gender Equality); and 6 (Clean Water and Sanitation), remain underexplored at the local level. This study addresses these gaps by providing empirical evidence on the relationship between freshwater scarcity and rural livelihood, highlighting vulnerability, and documenting household coping strategies in Jirapa Municipality of Ghana. The study thus aims to assess the relationship between freshwater scarcity and rural livelihoods (agriculture, health, education, and socio-economic wellbeing), and analyse the coping strategies adopted by households in response to freshwater scarcity. Consequently, it contributes novel empirical evidence by linking patterns of water access regularity with livelihood outcomes at the household level in a semi-arid Ghanaian context. It further situates freshwater scarcity within the broader SDGs framework, particularly goals 1, 3, 5, 6, and 13.

2 Conceptual framework

The conceptual framework for this study was adapted from the Department for International Development (DFID) Sustainable Livelihoods Framework (SLF) as used by [25]. The SLF represents the factors that affect people's livelihoods in communities, as well as typical relationships that exist between the elements defined in the theoretical framework [30]. Being a people-centered framework, it sought to provide a checklist of issues and their linkages, focusing on major influences and processes, as well as emphasizing the multiple interactions between different factors that influence people's livelihoods [31].

Freshwater scarcity is explicitly conceptualised in this study as a key indicator of the vulnerability context within the SLF. It captures both biophysical stressors (e.g., rainfall variability, declining groundwater, and seasonal shortages) and access constraints (e.g., distance, reliability, and competition for water), which collectively shape households' exposure to risk and limit access to natural capital. In this sense, water scarcity functions as a critical pathway through which environmental change translates into livelihood vulnerability.

The SLF also informed the choice of research variables and analytical procedures. Variables were organized around four core framework components: vulnerability context (water availability, distance, and seasonality), livelihood assets (human, natural, and financial capital), livelihood outcomes (socioeconomic well-being, health, education, and cultural conditions), and coping strategies (e.g., rainwater harvesting, storage, and rationing). Descriptive statistics were used to characterize vulnerability and access patterns, while Likert-scale analysis collected felt impacts, and Pearson Chi-square tests looked at the relationship between water shortage and livelihood outcomes. Qualitative analysis expanded on these by describing context-specific experiences and institutional processes. This method assures that the empirical analysis is firmly rooted in the SLF.

The framework argues that range of livelihoods, including human, natural, social, financial and physical capital, are influenced by a set of structures and processes. The interaction leads to different livelihood strategies adopted by households to achieve a set of livelihood outcomes, including but not limited to increased income, well-being, reduced vulnerability and improved food security. For this study, the framework shows how water scarcity influences the livelihoods of people in small-holder farming communities in the Jirapa Municipality of Ghana's Upper West Region (see Fig. 1). Water is identified as a resource that households depend on for survival [14]. From the framework, the vulnerability context, which includes natural factors such as temperature, rainfall and siltation, is considered. Also, anthropogenic factors such as population pressure, pollution, deforestation, and mismanagement of water are recognized. These factors directly or indirectly influence households' access to the available livelihood resources (water). The form of influence of these factors together with the level of resource (water) management (e.g., technical planning and investment in water provisions, institutional legal framework such as water formal regulations and customary rules and norms) determines the water scarcity effects on livelihood aspects, including finance, poverty, health, education, culture, hunger, gender discrimination, among others [32]. Furthermore, it is perceived that the most vulnerable groups that bear the brunt of water scarcity are women and children [2]. Finally, households and communities under water scarcity conditions have to adapt strategies such as rainwater harvesting, reduce water use, dependence on surface water (rivers, ponds, dams, etc.), water pumping technology, and water vendor services to cope with the situation [33].

3 Methods and context of the study

3.1 Description of the study area

Jirapa Municipality forms part of the eleven Metropolitan, Municipal, and District Assemblies (MMDAs) of Ghana's Upper West Region. Jirapa, the municipal capital, is situated approximately 63 km north of Wa, the regional capital. Geographically, the

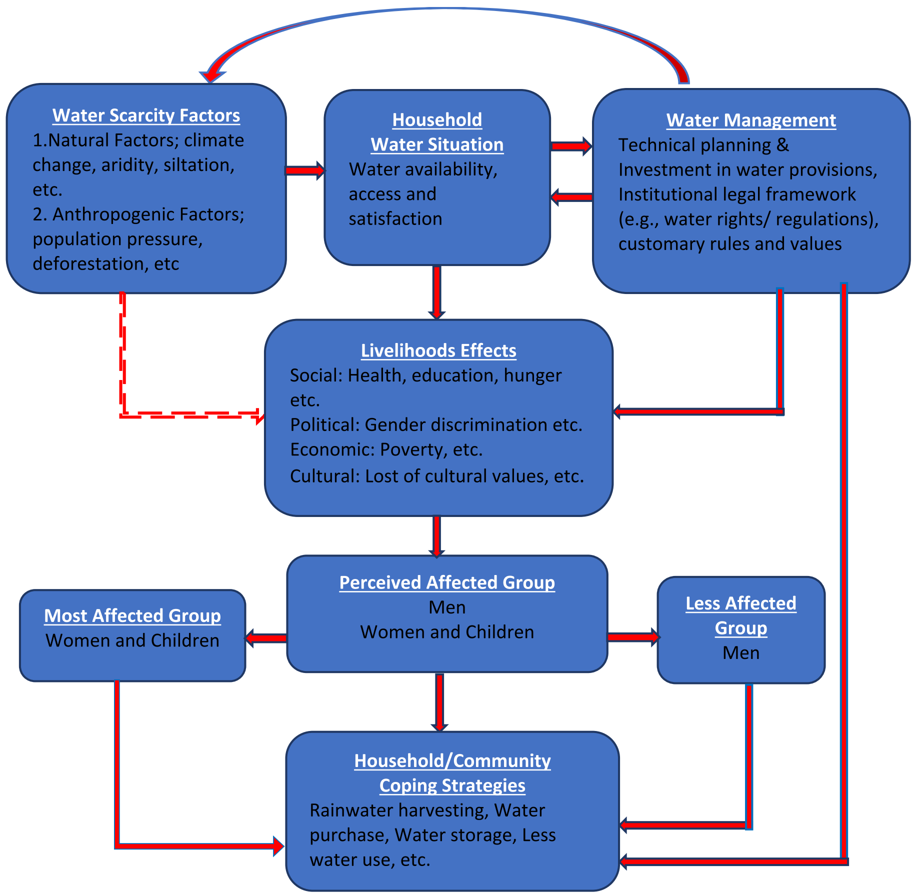

Fig. 1 Conceptual framework for freshwater scarcity. *Source* Adapted from [31]

municipality is located in the north-western corridor of the region, extending roughly between latitudes 10° 0.25″ and 11° 0.00″ North and longitudes 2° 0.25″ and 2° 0.40″ West, and covering an estimated land area of about 1188.6 km^2 [5]. Relatively, it shares boundaries with the Nadowli-Kaleo and Daffiama-Bussie-Issah districts to the south, Lambussie-Karni District to the north, Lawra Municipality to the north-west, Sissala West District to the east, and the Black Volta River forming its western boundary (see Fig. 2).

The Municipality is predominantly rural with livelihoods largely anchored in smallholder agriculture [5]. The agricultural production is primarily rain-fed, rendering livelihoods highly sensitive to climatic variability and environmental stressors [5, 11]. Hydrologically, Jirapa Municipality is located in the Black Volta basin, where seasonal streams and small, mostly transient tributaries dominate drainage networks. As a result, surface water supply is restricted and unpredictable, especially during the dry season [11, 45], hence the main supply of residential water is groundwater obtained through boreholes. Meanwhile, its dependability is limited by varying recharge rates and falling water levels in northern Ghana [9].

The municipality experiences a unimodal regime of rainfall, with annual totals ranging from about 800 to 1100 mm, mostly falling between May and October [5].

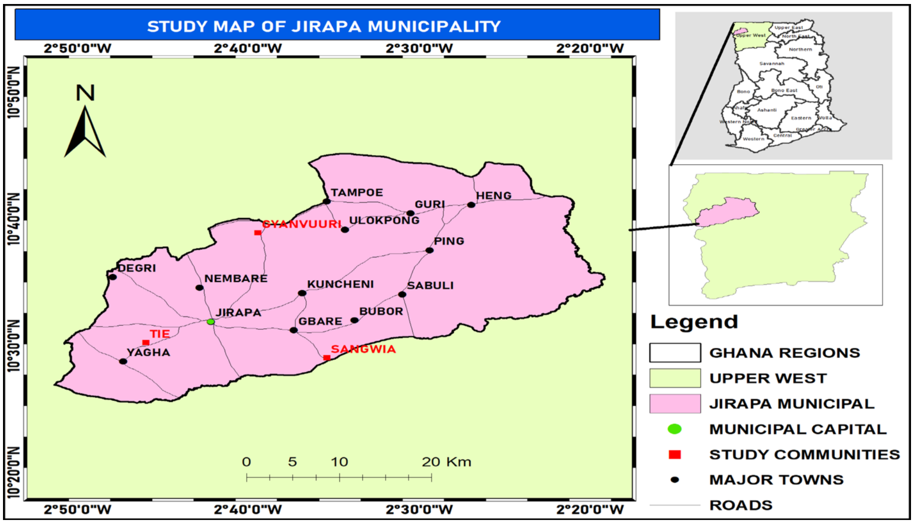

Fig. 2 Map of the study area. *Source* Authors' Construct based on GIS data 2025

However, rainfall patterns are becoming more and more characterized by high inter-annual variability, unpredictable initiation and cessation, and protracted intra-seasonal dry spells, which shorten the agricultural growing season and decrease effective groundwater recharging [7]. In semi-arid areas, these hydrological dynamics increase livelihood vulnerability and perpetuate seasonal water scarcity [3, 19]. Consistent with broader patterns observed across the sub-Saharan African region, where rural populations are largely dependent on small-scale farming systems [34], and more than 60% of Jirapa's population engages in smallholder agricultural activities [5, 11].

Jirapa Municipality was deliberately selected for this study due to its documented vulnerability to water scarcity and related environmental challenges [5]. Previous studies identify the Municipality as one of the water-stressed areas within the Upper West Region, where households experience persistent difficulties in accessing reliable and safe water sources [28]. These challenges are further compounded by increasing climate extremities, including prolonged dry spells and erratic rainfall patterns, which intensify pressure on already limited water sources [9]. The combination of high dependence on climate-sensitive livelihoods and chronic water constraints makes the municipality a relevant and critical setting for examining water-related challenges and their implications at the household and community levels.

3.2 Research design and sampling strategy

This study was guided by a pragmatic research philosophy, which prioritises the research problem and supports the integration of multiple methods to generate robust and actionable evidence, and accordingly supports the integration of qualitative and quantitative methods, overcoming the limitations of relying on a single approach [35]. To enhance methodological clarity, the study adopted a concurrent mixed-methods design, where quantitative and qualitative data were collected simultaneously, analysed separately, and integrated during interpretation. The integration of methods allowed for triangulation and validation of findings, thereby improving the reliability and depth of the results [36].

The sampling process combined both probability and non-probability techniques to ensure methodological rigour and relevance. The sampling strategy was implemented in two distinct phases. First, purposive sampling was used to select study communities and key informants based on their relevance, experience, and knowledge of water-related challenges. Second, a probability-based simple random sampling technique was applied to select household respondents from a defined sampling frame using the lottery method. This ensured representativeness and minimised sampling bias in the quantitative component of the study. In all, 152 household heads were randomly sampled from a total of 245 households across the three selected communities (including Tie, Sangwia and Gyanvuuri) in the Municipality, based on the most recent household data available [5]. The combined sampling strategy enhanced both the depth and generalisability of the findings. The sample size was determined using [38] formula: $n = \frac{N}{1+N(\alpha)2}$ where n = sample size, N = population size (245 households), and α = margin of error (0.05). The formula was expressed as $n = \frac{245}{1+245(0.05)2} = 151.9$, thus, n = 152.

The distribution of the survey sample across the three selected communities was determined using a proportional allocation approach. This method ensured that the number of households sampled from each community reflected its relative share of the total household population. Accordingly, the sample for each community was calculated using the formula [S = (sP/TP) * n] where S = sample size, sP = Population (households) of settlement (community), TP = total population (total households), and n = original sample size used for the study. The computations were done as follows: Tie = $\frac{113}{245} \times 152 = 70$; Sangwia = $\frac{64}{245} \times 152 = 40$; Gyanvuuri = $\frac{68}{245} \times 152 = 42$.

The sample size of 152 households was determined to ensure adequate representation of the study population while remaining feasible within logistical and resource constraints. The selection was guided by standard sampling considerations for social science research, where a sample exceeding 100 observations, is generally sufficient for basic inferential analysis, including chi-square tests [35]. Also, the sample size was proportionally distributed across the selected communities to capture spatial variations in water access and livelihood conditions, thereby enhancing the representativeness and reliability of the findings.

In addition to the 152 household heads who provided quantitative data, 20 respondents included nine (9) executives of farmer groups, six (6) community opinion leaders, and five (5) officials drawn from relevant institutions and the District Assembly served as key informants for the qualitative component of the study. These individuals were purposively selected based on their leadership, representative, and coordinating roles, which positioned them as knowledgeable sources on community-level water challenges and livelihood-related interventions. The participant types, institutional or community affiliations, as well as inclusion and exclusion criteria are listed in Table 1. Pseudonyms are employed to ensure that data sources can be traced across communities while maintaining secrecy.

The qualitative sample size was determined using the principle of data saturation, where data collection continued until no new information emerged from interviews and focus group discussions (FGDs). In addition, the sample size was considered adequate

Table 1 Selection and characteristics of key informants

Category	Source/institution	Sample size (n)	Inclusion criteria
Farmer group executives	Farmer based organizations	Tie = 3, Sangwia = 3 and Gyanvuuri = 3	Recognized leadership role (chairperson/secretary), ≥ 3 years farming experience, active participation in water-related livelihood activities
Community opinion leaders	Informal traditional actors	Tie = 2, Sangwia = 2 and Gyanvuuri = 2	Demonstrated knowledge of community water issues, leadership or advisory role, long-term residency (≥ 10 years)
Institutional actors	District assembly, water and sanitation agencies (e.g., CWSA), agriculture department	Municipality-wide (serving all study communities) = 5	Direct involvement in water governance, rural development, or agricultural extension services; minimum 2 years of professional experience

Source Authors' Construct, 2025

for capturing diverse perspectives across stakeholder groups (farmers, opinion leaders, and traditional authorities), consistent with established qualitative research standards which typically recommend 12–30 interviews for data saturation in homogeneous rural populations [36, 37].

Quantitative data were gathered using a structured questionnaire containing both closed- and open-ended items. Data collection was facilitated by two trained field assistants and administered digitally through the Kobo platform, which enhanced data accuracy and efficiency. Qualitative data were deepened using semi-structured interview guides that combined predetermined questions with broad thematic prompts to allow flexibility and depth during discussions. In total, five (5) key informant interviews comprising of institutional actors and two (2) FGDs made up of farmer group executives and community opinion leaders were conducted, with each session lasting for 45–60 min. The FGDs were organised separately for men and women, comprising eight male participants and seven female participants, respectively. Gender-segregated discussions were adopted to encourage free expression, particularly among women, who are often reluctant to voice their views in mixed-gender settings within the study communities.

The questionnaire captured data on socio-demographic characteristics, water access, distance to water sources, seasonal availability, livelihood impacts, and coping strategies. The qualitative instruments (mainly interview and FGD guides) focused on community perceptions, institutional challenges, gendered experiences and adaptive responses to water scarcity. The use of multiple instruments ensured complementarity and depth in data collection. In each community, both Key Informant Interviews (KIIs) and FGDs were conducted in designated neutral venues such as community centres or school compounds, depending on local availability and acceptability. Sessions were scheduled in consultation with the respondents. Interviews and discussions were conducted in the local language (Dagaare).

The study employed descriptive and inferential techniques consistent with the mixed-methods design [35]. Quantitative data were analysed using descriptive statistics (frequencies, percentages, and means) to summarise socio-demographic characteristics and water access patterns. The quantitative analysis was primarily conducted at the household level, and as such, most variables reflect aggregated household responses rather than individual-level gender-disaggregated data. This limits the extent to which statistical comparisons can be made between male and female respondents across all indicators. Inferential analysis was conducted using Pearson Chi-square (χ^2) tests to examine associations between water scarcity and livelihood outcomes. In addition, Likert-scale mean score analysis was applied to evaluate perception-based variables. Qualitative data were analysed using a descriptive narrative approach to complement and enrich the quantitative findings. The data analysis involved transcribing all handwritten notes and audio recordings into text to facilitate coherent interpretation and clear presentation of issues relevant to the study objectives. Interview and FGDs transcripts were reviewed in detail to identify relevant insights, recurring experiences, and illustrative examples related to water access, livelihood conditions, and coping strategies. The analysis focused on extracting key narratives and explanatory accounts that align with study objectives. These narratives were integrated directly into the results section to provide contextual understanding of statistical relationships and to capture participants' lived experiences.

To ensure accuracy, a structured transcription protocol was followed. Audio recordings were first transcribed verbatim by the lead researcher and independently reviewed by a trained research assistant. A back-checking procedure was employed, where selected transcripts were compared against audio recordings to verify meaning consistency. Where translation from Dagaare to English was required, a forward-translation and back-translation process was applied to minimise semantic distortion. This approach ensured that qualitative data served a complementary role, supporting the interpretation of quantitative findings within a convergent mixed-methods framework [35].

Quantitative data analysis was performed using the Statistical Package for the Social Sciences (SPSS), version 21.0. Descriptive statistics were employed to analyse respondents' socio-economic and demographic characteristics, freshwater scarcity indicators, and other key study variables. A five-point Likert scale (strongly disagree = 1 to strongly agree = 5) was used to gauge respondents' opinions of the effects of freshwater shortage on important livelihood variables.

Likert-scale items were analysed using frequencies, percentages, and mean scores. Although Likert responses are ordinal in nature, mean values were used as a measure of central tendency based on the common practice in social science research of treating Likert-type scales with multiple response categories as approximately interval-level data for descriptive purposes. This technique is commonly used in perception-based research and offers a solid foundation for comparing and evaluating subjective responses [39]. The combination of Likert-scale analysis with descriptive and inferential statistics strengthens the findings and supports qualitative insights. Results were presented using frequency tables, charts, and bar graphs. In addition, chi-square tests were conducted to examine relationships between the water scarcity situation and its relationship with socio-economic, livelihoods and cultural status among respondents.

The selection of analytical techniques was guided by the nature of the data and study objectives. Descriptive statistics addressed patterns of water access and socio-demographic characteristics, Likert-scale analysis captured perception-based vulnerabilities, and chi-square tests examined statistically significant associations between water scarcity and livelihood outcomes. Qualitative narrative analysis complemented these findings by providing contextual explanations, ensuring coherence between methods and results.

4 Results and discussions

4.1 Socio-demographic characteristics

To provide theoretical support, the findings are interpreted using the five livelihood assets of SLF: human, natural, financial, physical, and social capital. Freshwater scarcity, as proven in this study, largely limits natural capital (water availability), affecting other livelihood assets and outcomes. For example, limited availability of water affects agricultural output and revenue, eroding financial capital, but vast distances to water sources and poor infrastructure indicate physical capital limits. The increasing burden on women and children, as well as the implications for health and education, suggest further strains on human capital, whereas disruptions to social interactions and cultural traditions show social capital loss.

These interlinkages support the SLF proposition that shocks within the vulnerability context, such as water scarcity, cascade across multiple livelihood assets and shape household strategies and outcomes [30, 31]. The findings therefore demonstrate that freshwater scarcity is not an isolated environmental issue but a systemic constraint on livelihood systems, reinforcing multidimensional vulnerability in rural communities. This study examined key socio-demographic characteristics of respondents (see Table 2) to provide contextual insight into livelihood dynamics and vulnerability to freshwater scarcity. These characteristics are critical for understanding differential exposure, adaptive capacity, and resource-use behaviour in water-scarce rural settings (9, 40).

The findings show that 53% of the respondents were male and 47% were female. The sample shows a relatively balanced gender composition. In terms of age structure, 44% of the respondents were between 18 and 34 years, 34% fell within the 35–49-year category, and 22% were aged 50 years or above. Overall, more than two-thirds of the respondents (78%) were within the economically active age group of 18–49 years, indicating a predominantly youthful population. This pattern is consistent with national demographic trends, as the [5] report establishes that 56.8% of the population in Jirapa Municipality falls within the 15–64-year age group. A youthful and economically active population has important implications for water demand, labour availability, livelihood diversification, and sustained pressure on limited freshwater resources over time. Meanwhile, during the FGDs, the opinion leaders complained bitterly about the rampant north–south migration among the youth in the municipality. A 50-year-old man, for instance, vehemently lamented that " *….. in recent times, most of our active and energetic youth prefer travel to the down south to engage in other livelihood options, mainly 'galamsey,' following the challenges in climatic conditions, including unpredictable rainfall patterns and high temperature, nowadays.* This suggests that climate change resulting in freshwater shortages remains problematic in the study area, affecting the livelihoods of the rural folks, and hence the youth decisions to indulge in frequent down south migration and most at times the prohibited activities, including 'galamsey' and other social vices [16, 29].

Table 2 Socio-demographic characteristics

Variable	Freq (%)	Variable	Freq (%)
Sex		Household size	
Male	81 (53)	1–3	25 (16)
Female	71 (47)	4–6	53 (35)
Total	152 (100)	7–10	38 (25)
Age (yrs.)		11+	36 (24)
18–34	67 (44)	Total	152 (100)
35–49	52 (34)	Religion	
50+	33 (22)	Christian	104 (68)
Total	152 (100)	Islam	2 (1)
Marital status		Traditional	46 (30)
Married	114 (75)	Total	152 (100)
Single	25 (16)	Duration of stay (yrs.)	
Divorced	1 (1)	0–10	16 (11)
Widow/widower	12 (8)	11–20	22 (14)
Total	152 (100)	21 years and above	114 (75)
Educational level		Total	152 (100)
No formal education	69 (45)	Major occupation	
Primary	34 (22)	Farming	140 (92)
JHS/middle school	32 (21)	Trading	5 (3)
SHS/Tech/Voc	9 (6)	Civil/public service	6 (4)
Tertiary	8 (5)	Artisan	1 (1)
Total	152 (100)	Total	152 (100)

Source Field Survey, 2025

Regarding marital status, the majority of the respondents (75%) were married, while 16% were single, 1% divorced, and 8% widowed. The predominance of married respondents suggests that family-based household structures are common in the study area. Such arrangements are often associated with larger household sizes, which in water-scarce environments can substantially increase domestic water demand and intensify pressure on already constrained water resources. Household size further reinforces this pattern. Only 16% of the respondents reported small households of 1–3 members, while 35% had household sizes of 4–6 members, 25% reported 7–10 members, and 24% lived in households comprising 11 or more members. Overall, 84% of the respondents resided in households with four or more members. Large household sizes significantly elevate daily freshwater demand and might heighten competition over scarce water resources at both household and community levels, with direct implications for livelihood sustainability and well-being. According to empirical studies, home size correlates positively with water use and might aggravate vulnerability in water-scarce regions. For example, [40] and [41] highlight that growing household sizes put substantial demand on water supplies in rural areas, especially where access is limited.

Education, as a key component of human capital development, plays a critical role in shaping livelihood options and the capacity for sustainable natural resource management [42]. The results indicate that 55% of the respondents had some level of formal education, including primary education (22%), Junior High or Middle School (21%), Senior High/Technical/Vocational education (6%), and tertiary education (5%). However, a considerable proportion of the respondents (45%) had no formal education. This relatively low level of educational attainment could limit awareness of improved water management practices, alternative livelihood opportunities, and adaptive strategies in the face of increasing water scarcity. The occupational structure of the study communities reflects a strong dependence on natural resources, particularly freshwater. Similar findings were reported by [40, 43] in Nigeria and in northern Tanzania, respectively. Education has been shown to increases environmental knowledge and decision-making ability, particularly in rural populations [44].

Farming emerged as the dominant livelihood activity, engaging 92% of the respondents. Other occupations included trading (3%), civil or public service employment (4%), and artisanal work (1%). This pattern aligns with the Municipal development reports that identify agriculture as the primary economic activity in Jirapa Municipality [11]. The heavy reliance on farming underscores the high sensitivity of local livelihoods to water availability and climate variability, highlighting the centrality of reliable freshwater resources to both household welfare and local economic stability [34, 44].

4.2 Main sources of freshwater and seasonal water scarcity

The findings indicate that groundwater, which is mainly accessed through boreholes, constitutes the primary source of freshwater for domestic use in the study communities. Thus, more than half of rural households (54.6%) depend principally on boreholes for their daily water requirements, while 22.4% rely on surface water sources such as dams, streams, and ponds, and 23% use rainwater as an alternative supply. This finding was confirmed during the interview sessions when an officer from CWSA said:

> *.... most of the rural residents rely on available but limited boreholes, that are sometimes located a bit far away in their communities, with only a few depending on surface and rain waters, perhaps due to their unreliability for many months within the year .... because of the long dry season which brings complete stoppage of the rains and total dryness of the surface water (A 38-year-old lady, KIIs, February, 2025).*

These patterns suggest that households' choice of domestic water sources is driven largely by physical availability and ease of access rather than quality or reliability considerations. The results are also consistent with municipal records showing that boreholes remain the dominant source of domestic water across communities in the Jirapa Municipality and that smallholder farming communities have virtually no access to pipe-borne water systems [11]. Again, it aligns with studies in rural Ghana that show a high reliance on groundwater due to insufficient infrastructure and unequal access to upgraded water systems [6, 8].

Notwithstanding their central role in domestic water provision, these sources are widely perceived as unreliable and unsustainable, particularly during the dry season. Qualitative evidence from FGDs underscores persistent challenges, including frequent borehole breakdowns, low discharge rates, and prolonged queuing, especially in situations where several community sections depend on a single borehole. A 54-year-old woman, during a FGD in Tie, indicated that *"boreholes often yield little or no water during the dry season, compelling residents to spend long hours waiting for minimal quantities of water".* Such accounts vividly illustrate the daily struggles associated with accessing domestic water and highlight the vulnerability of households to intermittent supply. These conditions limit households' ability to secure adequate water for daily use and livelihoods, highlighting the precarious nature of water access. Existing water sources fail to meet household demand, exposing residents to chronic water scarcity. The reported unreliability of water sources corroborates existing evidence that rural water systems in sub-Saharan Africa are often characterised by seasonal variability, infrastructure breakdowns, and limited sustainability [9, 28].

In contrast, agricultural water use in the study area is overwhelmingly dependent on natural rainfall. Approximately 85% of respondents indicate relying exclusively on rainfall for their crop production, while only small proportions use groundwater (3%) or surface water (12%) for limited dry-season gardening. This pattern is consistent with municipal assessments that identify farming in the Jirapa Municipality as predominantly rain-fed, with minimal irrigation development [11]. During the interview sessions, an extension officer from the Agriculture Department confirmed the situation by explaining that

> *....... almost all the farmers in the municipality are vulnerable to the climate change challenges .... majority of them depend solely on rainfall for all their farming activities, so continuously, if the rain fails, their crop yields also fail.... this situation not only threatens the farmers' livelihoods but makes them remain poor (A 40-year-old man, KIIs, February, 2025).*

This finding reflects wider regional trends in sub-Saharan Africa, where more than 95% of arable farming depends on rainfall [20, 45]. However, the surface and groundwater sources used for supplementary irrigation are themselves unreliable, as they often dry up during extended dry periods due to climatic variability and declining water tables. A 65-year-old male farmer noted, during the FGDs, that:

> *........ farming activities are virtually impossible outside the rainy season due to the absence of dams and reliable irrigation infrastructure. Once the rains stop, crop production ceases because available surface water sources dry up rapidly and alternative water sources for irrigation are either unavailable or insufficient (Male FGDs, February 2025).*

These narratives reinforce the broader quantitative findings and demonstrate how inadequate and unreliable water sources undermine both household welfare and productive activities. This trend is largely congruent with regional and global research that suggests that smallholder agriculture in Sub-Saharan Africa is mostly rain-fed and very sensitive to rainfall variability [19, 45].

The study also demonstrates that freshwater scarcity in the study communities follows a pronounced seasonal pattern (see Fig. 3). According to 74% of the respondents, water shortages are most severe between January and May, a period characterised by minimal or no rainfall, high temperatures, and elevated evaporation rates. These conditions accelerate the drying of surface water sources and reduce borehole yields as groundwater tables decline, intensifying difficulties in accessing freshwater for both domestic and agricultural purposes. In contrast, water scarcity is least pronounced between June and September, as indicated by only 8% of the respondents. This period coincides with peak

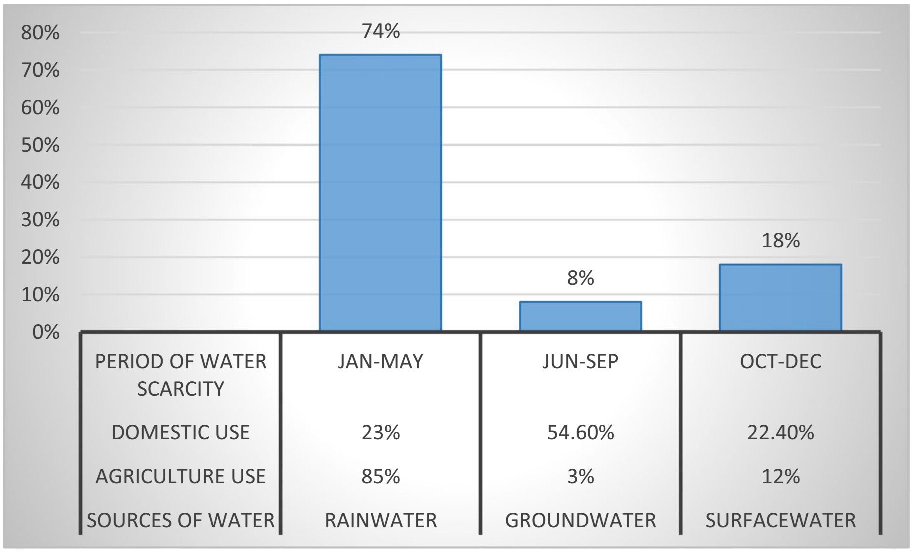

Fig. 3 Major sources of freshwater and seasonal water scarcity. *Source* Field Survey, 2025

rainfall, lower temperatures, and reduced evaporation, during which households commonly engage in rainwater harvesting, thereby easing pressure on boreholes. Reservoirs and surface water bodies are also replenished, facilitating easier access to freshwater for various non-potable uses. However, after September, water shortages begin to intensify again, as reflected by 18% of the responses, due to declining precipitation and the onset of unfavourable climatic conditions. The seasonal character of water scarcity seen in this study is consistent with climatological and hydrological data for northern Ghana, where rainfall variability causes cyclical water availability [45].

4.3 Distance covered by rural communities to access freshwater

The distance and time required to reach water sources serve as key indicators of the state of freshwater availability in rural communities [2]. Shorter distances improve access, while longer distances increase effort and time costs. The findings reveal that 74% of the households travel between 500 m and 1 km or more to access water for domestic purposes (see Fig. 4). This reflects significant constraints in freshwater access rather than absolute water scarcity alone. This journey translates to approximately 30 min of walking, not accounting for additional time spent queuing for water, particularly under conditions of high demand and overcrowding. Only 26% of households have relatively proximate access, covering distances between 100 and 500 m.

These results are consistent with the observations of [7], who note that, for many rural households, the nearest source of domestic water typically requires at least a 30-min walk. The extended distances observed in the study communities could be attributed largely to the dispersed settlement pattern characteristic of rural areas in the municipality. In his explanations, a local Assemblyman remarked:

> *The settlements of most rural communities in the municipality are so dispersed with small nucleated sections that are distant apart, making it difficult for balanced distribution of artificial freshwater supply, such as boreholes and reservoirs. This*

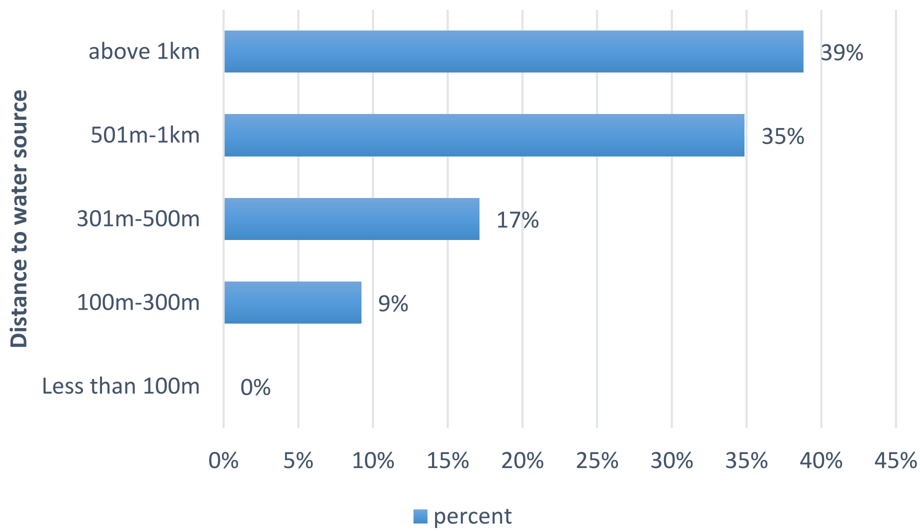

Fig. 4 Distance covered to freshwater sources. *Source* Field Survey, 2025

> *results in water supply challenges in many rural communities where people have to walk long distances to access freshwater for their households (A 32-year-old Assemblyman, KII, February 2025).*

The issue of distance to the available freshwater sources came out strongly during the FGDs in almost all the communities, as a good number of participants in both female and male sessions expressed their displeasure. A participant, for instance, lamented that:

> *…in fact, most of us, I mean, we the women, are not happy at all living in our current communities because of freshwater availability and accessibility. Tellingly, sir, do you know that we always have to walk a considerable distance before getting access to freshwater for our household use? This situation demotivates us from even engaging in any economic activities that use more freshwater (A 45-year-old woman from Sangwia, female FGDs, February 2025).*

The implications of these long distances and extended collection times are significant. Nolan [16], for instance, observes that such conditions disproportionately affect women and girls, limiting school attendance for girls and constraining women's capacity to engage in household management, income-generating activities, and other productive labour. Consequently, distance and time requirements not only reflect the physical burden of accessing freshwater but also have broader socio-economic and developmental consequences for rural households. The disproportionate burden on women and girls has been well documented in the literature, highlighting the gendered nature of water access and its implications for education and economic participation [17, 21].

4.4 Livelihoods associations with fresh water scarcity

4.4.1 Water scarcity association between socio-economic and cultural livelihoods

The cross-tabulation and Pearson Chi-square test findings show a robust and statistically significant relationship between freshwater scarcity and household livelihood outcomes in the research region (see Table 3). Water scarcity is significantly associated with socio-economic class ($\chi^2(2) = 21.073$, $p = 0.001$). Notably, 91.3% of the respondents facing severe water shortage indicated bad socio-economic livelihood outcomes, whereas just 14% reported poor livelihoods in non-severe water scarcity situations, underscoring the extent of vulnerability among impacted households. In a similar vein, there was a significant correlation ($\chi^2(2) = 29.949$, $p = 0.001$) between freshwater scarcity and cultural status, with 86% of the respondents experiencing severe water scarcity indicating deteriorating cultural conditions, reflecting disruptions to social relations, cultural practices, household cohesion, and community participation. An interview with a 35-year-old officer from the office of rural development unveiled that water scarcity disrupts not only the livelihoods of the rural folks, but their cultural practices as well, when he vehemently lamented that "*hmmmm, because of water scarcity we cannot even enjoy celebrating most of our festivities, such as funerals, naming ceremonies and even our annual ritual/festival*". These findings are consistent with empirical studies demonstrating that water scarcity is significantly associated with both economic productivity and socio-cultural cohesion in vulnerable rural communities [22, 44]. The widespread social repercussions of chronic water insecurity are shown by the fact that only a tiny percentage of respondents under severe water scarcity indicated positive or average cultural results.

These findings show that freshwater scarcity is significantly associated with livelihood outcomes. Globally, an estimated 2.3 billion people are now living in water-scarce situations [1], indicating a growing imbalance with far-reaching consequences for livelihood sustainability. This mismatch offers especially serious implications in arid and semi-arid countries, where freshwater supply is inextricably linked to socioeconomic production systems and cultural practices. The magnitude of these observed associations is consistent with global estimates of increasing water scarcity, highlighting the systemic nature of water insecurity as a development challenge [3].

4.5 Water scarcity relationship with agricultural production, health, sanitation and children's education

Beyond aggregate livelihood indicators, freshwater scarcity exerts direct pressure on household economic activities, food systems, and human development outcomes.

Table 3 Water scarcity association with socio-economic and cultural livelihoods

Variable		Water scarcity situation/condition		Total
		Not severe	Severe	
Socio-economic and livelihood status	Good	5 (3.3%)	5 (3.3%)	10 (6.6%)
	Average	6 (3.9%)	6 (3.9%)	12 (7.9%)
	Poor	14 (9.2%)	116 (76.3%)	130 (85.5%)
Total		25 (16.4%)	127 (83.6%)	152 (100%)
N = 152, $\chi^2(2) = 21.07$, $p = 0.001$				
Cultural status	Good	5 (3.3%)	5 (3.3%)	10 (6.6%)
	Average	6 (3.9%)	14 (9.2%)	20 (13.2%)
	Poor	5 (3.3%)	117 (77%)	122 (80.3%)
Total		16 (10.5%)	136 (89.5%)	152 (100%)

N = 152, $\chi^2(2) = 29.95$, $p = 0.001$

Source Field Survey, 2025

Likert-scale analysis ranging from 1 to 5 (see Table 4) reveals strong consensus that water scarcity is associated with poverty and hunger (mean = 4.84) by constraining income-generating activities such as dry-season irrigation, pito brewing, gardening, and livestock production. A middle-aged woman in Tie sharing her view during a FGD said:

> *We keep struggling with poverty and hunger here because of water problems. This place as soon as it's dry season, it becomes very difficult to do any serious work that has to do with water. Even pito brewing, which is one of the commonest livelihood activities of women here, many women have stopped because the stress in getting water for that is not a joke (Female FGDs, February 2025).*

Respondents also strongly agreed that declining and erratic rainfall reduces crop yields (mean = 4.73) and undermines livestock survival (mean = 4.74), reinforcing cycles of seasonal unemployment and distress migration. Another respondent, a 70-year-old man, also stressed on the matter during an FGD, adding that:

> *These days the rainfall pattern is very bad, affecting our crop yields, but we also don't have a dam that could enable us to do irrigation during the dry season. Because of that, most of the young people usually travel to southern Ghana during that period to look for money (Male FGDs, February 2025).*

These findings align with earlier studies in Ghana and comparable contexts, which link rainfall variability and water scarcity leads to declining agricultural productivity and food insecurity [19, 25].

Table 4 Perceived relationship of freshwater scarcity on agricultural production, health, sanitation and children's education

Perception statements	SA	A	N	D	SD	Mean score
	Case counts (%)					
Water scarcity contributes to the reduction in agricultural output due to erratic rainfall	115 (75.7)	33 (21.7)	4 (2.6)	–	–	4.73
We usually lose many of our animals during dry season because they stray to far places in search of water and end up being stolen or killed	114 (75.0)	37 (24.3)	1 (0.7)	–	–	4.74
Water scarcity contributes to increasing poverty and hunger in the community (especially among women and children) as it limits various sources of income generation e.g. irrigation/dry-land farming, pito brewing, etc.	128 (84.2)	24 (15.8)	–	–	–	4.84
Water scarcity increases women's burden in the household and the community	108 (71.1)	39 (25.7)	1 (0.7)	4 (2.6)	–	4.65
Water scarcity limits access to safe water for drinking and practicing basic hygiene at home and in school, thereby affecting the health of people	83 (54.6)	69 (45.4)	–	–	–	4.55
Our children (especially the girls) go to school late and tired after spending long hours looking for water	58 (38.2)	80 (52.6)	13 (8.6)	1 (0.7)	–	4.28
Our children perform poorly in school because they usually lose concentration in class, and don't get enough time to study at home after spending much time each day to get water	23 (15.1)	104 (68.4)	24 (15.8)	1 (0.7)	–	3.98

SA Strongly Agree (5), *A* Agree (4), *N* Neutral (3), *D* Disagree (2) and *SD* Strongly Disagree (1)

Source Field Survey, 2025

Freshwater scarcity also poses serious threats to health, sanitation, and education. Over half of the respondents (54.6%) strongly affirmed that water shortages limit access to safe drinking water and basic hygiene, forcing households to rely on unsafe sources and reduce water use for bathing and washing. A 42-year-old woman in Gyanvuuri expressed her views regarding the situation in a FGDs that:

> *The water situation here is very bad. When it gets to the dry season, it's always a big problem. When it happens that way, bathing and washing of clothes become secondary matters, which is not good for our health. That is when people sometimes get all kinds of diseases and sicknesses (Female FGDs, February 2025).*

These coping behaviours increase vulnerability to water-borne diseases such as diarrhoea and typhoid, corroborating earlier findings by [1].

Similarly, water scarcity disrupts children's education, particularly for girls. The majority of respondents reported late school attendance, fatigue, and reduced academic concentration due to time spent hauling water. A 48-year-old mother in Gyanvuuri noted in a FGD that:

> *Many a time, in the dry season, mothers have to wake the grown-up girls up early every morning and together go to look for water before they go to school. In some cases, the boys also go to carry water but with bicycles. This can sometimes make the children go to school late, and at times, they will not even go at all. This usually affects their studies and performance in school (Female FGDs, February 2025).*

This supports the assertions of [17, 21] that water scarcity disproportionately undermines girls' educational outcomes, with long-term implications for human capital development.

4.6 Relationship of water scarcity on gendered burdens, household relations, and cultural disruptions

Freshwater scarcity also reconfigures gender roles, household dynamics, and cultural practices in ways that deepen social vulnerability (see Table 5). More than 70% of the respondents strongly agreed that water scarcity leads to an increase in women's workload, as women and girls remain primarily responsible for water collection in the household. This contributes to physical exhaustion and also heightens domestic tensions and conflict, as confirmed by 61.2% of the respondents. Two respondents, during FGDs, expressed their views on the matter as follows:

> *The water situation in this community is making young women to grow old by force because of the burden and struggle we go through every day for water and other chores at home. Anywhere you go and whatever you're doing but you're always thinking about water (A 47-year-old woman, FGDs, February 2025).*

Regarding water scarcity-induced household conflict, a 56-year-old man in Tie said:

> *As for water issues, there will always be quarrels at home, especially during the dry seasons when the water situation is very hard. You know, that time both human beings and animals are usually competing for water. Just imagine your wife struggles*

Table 5 Perceived relationship of freshwater scarcity on gendered burdens, household relations, and cultural disruptions

Perception statements	SA	A	N	D	SD	Mean score
Water scarcity affects conjugal relationship of couples, as wives usually wake up very early at dawn and use the time their husbands need them most in bed to search for water and return late	46 (30.3)	100 (65.8)	6 (3.9)	–	–	4.26
Water scarcity sometimes leads to household conflicts and that of the community as well	93 (61.2)	58 (38.2)	1 (0.7)	–	–	4.61
Sometimes women from other places don't want to marry to this community because of water scarcity	103 (67.8)	49 (32.2)	–	–	–	4.68
We usually struggle when building a house here because of water problem, as we have to travel far to get water	106 (69.7)	41 (27.0)	4 (2.6)	1 (0.7)	–	4.66
Water scarcity reduces the effectiveness of many of our social-cultural activities like funerals, festivals and local weddings	18 (11.8)	84 (55.3)	44 (28.9)	4 (2.6)	2 (1.3)	3.74

SA Strongly Agree (5), *A* Agree (4), *N* Neutral (3), *D* Disagree (2) and *SD* Strongly Disagree (1)

Source Field Survey, 2025

to get a few buckets of water for the household's use, and you, the man, end up emptying that for your animals to drink; you see the kind of confusion that will cause (Male FGDs, February 2025).

These findings are consistent with [22], who documents water-related conflict at both household and community levels in water-scarce regions.

The social consequences extend into marriage and family life. A strong majority of the respondents (mean = 4.68) indicated that women from neighbouring communities are increasingly reluctant to marry into water-scarce areas, while some existing marriages keep on experiencing strain due to time conflicts, fatigue, and reduced intimacy associated with water collection responsibilities. A 38-year-old man in Tie recounted his experience during an FGD as:

…several times, marriage proposals of young guys from this place have been turned down because of the water problem. The ladies will tell you they don't want to come and suffer. Sometimes, they will even come but at last, some do run away (Male FGDs, February 2025).

Also, a female respondent of age 41 said;

This water issue sometimes spoils people's relationships. The time your husband may want to have intimacy with you, that is when you will leave him alone on the bed and go to look for water. If care is not taken, you will come back and meet your bag and clothes waiting for you outside (Female FGDs, February 2025).

These findings do not deviate from what has been established in the literature. Thus, comparable patterns have been reported in Ghana and India, where chronic water shortages influence marriage decisions and marital stability [21].

Finally, freshwater scarcity disrupts housing construction and socio-cultural practices. Nearly 70% of the respondents reported that limited water availability delays or prevents housing construction, particularly during the dry season when labour is available but water is scarce. Cultural events such as funerals, festivals, and weddings are also affected, as water demand during such gatherings exceeds supply, leading to compromised hygiene and shortened ceremonies. A 46-year-old man resident in Tie shared his view during FGDs saying:

> *We suffer a lot when we want to build a house, and you know, constructing a house takes a lot of water, but we don't have any reliable source of water here. Getting water is always a big problem; you have to hire a tricycle to fetch freshwater from somewhere for you. Which means if you cannot afford the cost of the tricycle, your work cannot be done (Male FGDs, February 2025).*

A 68-year-old woman further lamented over the situation during FGDs:

> *I wish you were always here to see how people struggle for water when some of these events take place, it's very bad. Some people don't bath until the whole event is over, especially when it is a funeral. Because of that, these days we try to shorten every such event to reduce the burden of an unhygienic situation (Female FGDs, February 2025).*

These findings reinforce the argument that water insecurity undermines not only daily survival but also the social and cultural fabric of rural communities [12, 13].

4.7 Household coping strategies to water scarcity

Water scarcity forces households to adopt coping strategies to sustain daily life. In the smallholder farming communities of the Jirapa Municipality, the study identified multiple adaptive strategies that households employ to manage limited freshwater supply, including rainwater harvesting, reduction of water use, reliance on water vendors, and water storage.

Rainwater harvesting emerges as a primary coping strategy in the study communities, particularly during the rainy season (see Fig. 5). Households routinely collect rainwater from rooftops into containers such as basins, barrels, and tanks to supplement domestic water needs. The survey results indicate universal adoption of this strategy, with 100% of the respondents reporting rainwater harvesting as a mechanism for securing water during periods of scarcity. Qualitative evidence reinforces this finding as a 38-year-old mother in a FGD stated: *"When it is rainy season, we're always saved from water problems because we don't struggle to get water again. People harvest rainwater every time and store it for use; you only go to the borehole occasionally for drinking water"*. This observation aligns with [25], who identifies rainwater harvesting as a widely adopted coping strategy in water-scarce rural communities in Ghana.

During the dry season, when water scarcity intensifies, households often reduce the quantity and frequency of water use as a coping mechanism. Survey data indicate that 97% of the respondents employ this strategy, modifying domestic routines such as bathing, washing clothes, and other water-intensive activities. FGDs sections reveal practical adaptations including households reducing bathing from twice to once daily and cutting water volumes by half to conserve the limited supply. A 34-year-old mother emotionally stated that *"In the dry season, if you don't manage water properly, you will always have problems......we reduce the quantity of water we use for bathing*

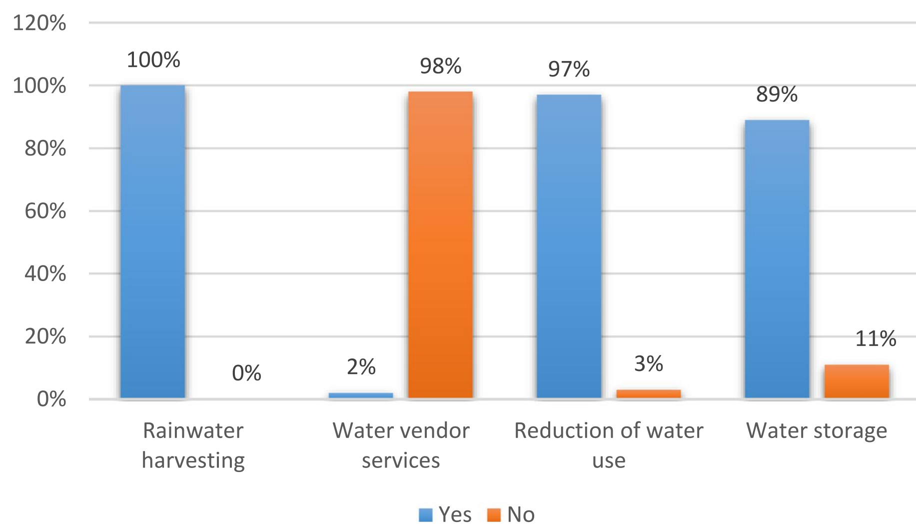

Fig. 5 Household coping strategies to water scarcity. *Source* Field Survey, 2025

and washing. If we don't, the situation becomes worse". Such practices are consistent with [24], who observes that households including nursing mothers constrain water use to maintain access to limited supplies, highlighting the behavioural adaptations necessary under water scarcity.

The study also examined the extent to which households purchase water from private vendors as a coping strategy. Only 2% of the respondents reported relying on water vendors, primarily individuals engaged in local production or construction activities. The overwhelming majority (98%) do not utilise this option, reflecting the rural and dispersed nature of the study communities, which limits the feasibility of commercially supplied water. This finding corroborates [33], who notes that water vending is more common in urban than rural contexts.

Water storage constitutes another critical strategy for mitigating the effects of unreliable supply. The study found that 89% of the households engage in storing water in containers such as drums, gallons, large earthen pots, and basins, enabling short-term availability for domestic use. The storage durations are, however, limited, typically lasting between three days and one week, due to the small size of storage containers, lack of technical know-how, and financial constraints. No households reported long-term water storage systems or storing water for agricultural purposes, citing high costs and technical barriers. In her explanation to the situation, a 42-year-old mother, during the FGDs, indicated:

> *....because the water sources are far, we fetch water to last about three days before going again......depending on container size and household needs, some people fetch water twice a day, so long-term storage is not feasible". Another participant remarked that "We don't have the capacity to store rainwater for agricultural purposes...constructing a dam would help, but we cannot do that ourselves (Female FGDs, February 2025).*

These relations consolidate [4] assertion that large-scale water storage for irrigation and dry-season agriculture is technically and financially demanding, explaining why smallholder farmers in resource-poor communities are unable to implement such strategies.

5 Conclusion

This study examined the livelihood relations of freshwater scarcity and the coping strategies adopted by rural households in Jirapa Municipality. The findings demonstrate that freshwater scarcity is a pervasive and multidimensional challenge with far-reaching implications for rural livelihoods. Households are highly dependent on groundwater sources, particularly boreholes (54.6%), yet these sources are unreliable, with 74% of respondents experiencing severe seasonal water scarcity between January and May. The burden of access is substantial, as approximately 74% of households travel more than 500 m to obtain water, resulting in significant time and labour costs, particularly for women and children.

The findings indicate a strong association between freshwater scarcity and adverse livelihood conditions. A large proportion of households experiencing severe water scarcity also reported poor socio-economic conditions (over 91%) and declining cultural well-being (86%). Perception-based results further suggest that respondents widely associate water scarcity with increased poverty and hunger (mean = 4.84), reduced agricultural productivity, threats to livestock survival, and challenges related to health, sanitation, and children's education. In addition, qualitative narratives highlight gender-differentiated experiences, particularly increased workload for women, heightened household pressures, and disruptions to social and cultural practices.

Although households adopt multiple coping strategies such as rainwater harvesting (100%), water rationing (97%), and short-term storage (89%), these measures are largely reactive, short-term, and insufficient to build long-term resilience. The absence of sustainable water infrastructure, particularly for dry-season farming and domestic supply, further exacerbates vulnerability. The study concludes that freshwater scarcity is strongly associated with livelihood challenges, including poverty, gender inequality, and socio-economic vulnerability. Addressing this requires integrated, context-specific interventions, including improved water infrastructure and livelihood diversification.

6 Recommendations

By empirically linking water access regularity, livelihood impacts, and coping strategies within a mixed-methods framework, the study advances existing scholarship on rural water insecurity in Ghanaian rural communities. The findings thus highlight the need for contextually appropriate and locally responsive interventions to address freshwater scarcity and its impacts on rural livelihoods. While the results demonstrate significant effects on agriculture, income, health, and education, the study does not directly assess institutional performance or governance structures. Therefore, rather than prescribing specific institutional actions, the study suggests that future policy efforts should consider strengthening water access, improving local water infrastructure, and supporting community-level adaptation strategies. In particular, interventions that enhance water availability during dry seasons and reduce the burden on vulnerable groups, especially women and children, who are likely to yield meaningful livelihood benefits. It is further recommended that a study be conducted to assess the effectiveness of policy approaches/strategies employed in the supply management of community water. Further studies should also focus on the effectiveness of policy approaches and/or strategies for ensuring sustainable water supply in rural communities.

7 Limitations of the study

This study has some limitations that should be acknowledged. First, the reliance on self-reported (perception-based) data, particularly through Likert-scale responses, may be subject to response bias. Second, the study focused on selected communities within the municipality, and therefore spatial variations across the entire municipality may not be fully captured to ensure the generalisability of the findings beyond those study communities. Third, the study involved community opinion leaders in FGDs, these participants were non-hierarchical actors, and their inclusion was not expected to significantly influence group dynamics. However, future studies may further disaggregate participant categories to enhance analytical depth.

A fourth limitation of this study is the limited gender-disaggregated quantitative analysis. Since the survey data were largely collected at the household level, it was not possible to fully examine gender differences across all livelihood indicators using statistical methods. Future research should incorporate individual-level data collection and gender-disaggregated analysis to better capture these dynamics. Finally, the quantitative analysis was primarily descriptive and associative in nature, and did not incorporate advanced modelling techniques, which could further deepen causal inference. Despite these limitations, the use of a mixed-methods approach enhances the robustness of the findings through triangulation.

Author contributions

B.Y.B, J. A, and F.S, Conceived the idea, Collected Data, Analyzed the data, Drafted the original manuscript, Edit the manuscript.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Ethical approval was obtained from the Institutional Review Board (IRB) of the University of Business and Integrated Development Studies formally known as the Simon Diedong Dombo University of Business and Integrated Development Studies, Wa, Ghana in accordance their ethical guidelines and protocols. The study was then carried out with strict adherence to the ethical guidelines and protocols of the Institutional Review Board (IRB) of the University of Business and Integrated Development Studies.

Verbal Informed consent was obtained from all study participants, ensuring they were briefed on the objectives and purpose of the research. They were also assured of confidentiality and made to understand participation was strictly voluntary, with the option to withdraw at any stage without consequence.

To ensure privacy and anonymity, no personally identifiable information was recorded, and pseudonyms were used in all transcripts and reporting. Furthermore, all data were treated with strict confidentiality, with access limited to the research team and securely stored for research purposes only. During the FGDs, participants were also informed of the need to respect the confidentiality of shared information within the group setting.

Consent for publication

Not applicable.

Competing interests
The authors declare no competing interests.

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